Inference often combines conditioning and summarizing

Observed		**Condition**		**Summarize**
y	$\rightarrow$	$p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z\mid y}, \Sigma_{z\mid y}\right)$	$\rightarrow$	$\hat{z} = E[z \mid y] = \mu_{z\mid y}$

Murphy, Book 2 §2.3.2.2–2.3.2.3

$$E[z \mid y] = \int z p(z \mid y)\, \mathrm{d}z$$

- Conditioning preserves the whole posterior distribution and its uncertainty
- Summarizing extracts a quantity from that posterior; for a Gaussian the mean is $\mu_{z\mid y}$

Choose a reconstruction by its conditional squared-error risk

Squared-error risk

Observed y; $z, a \in \mathbb{R}^d$

$$E[\|z\|^2|y] < \infty$$

$$L(a, z) = \|z - a\|^2$$

$$R(a|y) = E[L(a, z)|y]$$

Centered error

$$m = E[z|y], \quad r = z - m$$

$$z - a = r + (m - a)$$

Squared-error expansion

$$\|z - a\|^2 = \|r\|^2 + 2r^T(m - a) + \|m - a\|^2$$

- Choose a to minimize conditional expected squared error $R(a|y)$.
- Given y, offset $m - a$ is fixed; uncertainty remains in r.

Centering separates residual uncertainty from estimation error

Conditional expansion Previous slide: pointwise squared-error expansion

$$\|z-a\|^2 = \|r\|^2 + 2r^T(m-a) + \|m-a\|^2$$

Take $E[\cdot\,|y]$; $m-a$ is fixed

Cross-term

$$E[r|y] = E[z|y] - m = 0$$
$$E\left[r^T(m-a)|y\right] = E[r|y]^T(m-a) = 0$$

Residual energy

$$V = \mathrm{Cov}[z|y]$$
$$E\left[\|r\|^2|y\right] = \sum_{j=1}^{d} E\left[r_j^2|y\right] = \mathrm{tr}\, V$$

$$R(a|y) = \mathrm{tr}\, V + \|m-a\|^2$$

Murphy, Book 1 §5.1.5.1; vector decomposition: course elaboration

- Risk is residual uncertainty plus squared distance from the posterior mean.

The posterior mean minimizes squared-error risk

$$R(a|y) = \underbrace{\operatorname{tr} V}_{\text{independent of a}} + \underbrace{\|m-a\|^2}_{\text{nonnegative; zero iff } a=m}$$

Consequences at the minimizing action

Minimum risk

$$\min_a R(a|y) = \operatorname{tr} V$$

Optimal action

$$a^*(y) = m = E[z|y]$$

For the conditional Gaussian

$$\min_a R(a|y) = \operatorname{tr} \Sigma_{z|y} \qquad a^*(y) = \mu_{z|y}$$

- The conditional mean minimizes model expected squared error.
- The posterior covariance describes uncertainty that remains after the mean action is chosen.

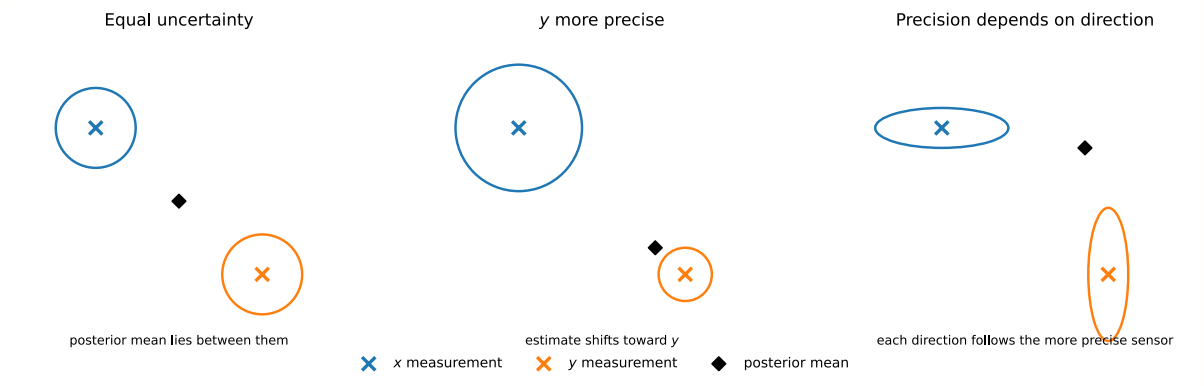
Sensor fusion combines measurements according to their uncertainty

Two noisy measurements of one state

$$z \sim \mathcal{N}(\mu_z, \Sigma_z)\; x \mid z \sim \mathcal{N}(z, \Sigma_x), \quad y \mid z \sim \mathcal{N}(z, \Sigma_y)$$

Same two operations

$$p(z \mid x, y) \quad \rightarrow \quad E[z \mid x, y]$$



Qualitative weak-prior-limit schematic after Murphy, Book 2 §2.3.2.3, Fig. 2.10; `sensor_fusion_figure.ipynb`

- With independent sensor noises, equal isotropic noise gives the midpoint in the weak-prior limit; finite-prior mean and precision also matter.
- A more precise sensor pulls the posterior mean closer to its measurement
- With anisotropic uncertainty, the weighting can differ by direction